

■ Plan, Feasibility, and Key Risks

Plan

- ▶ **Algo Type1:** LOB-powered execution layer.
- ▶ **Internal flow:** crossing, routing, and liquidity timing.
- ▶ Value metrics: slippage, impact, fill quality, internalization.

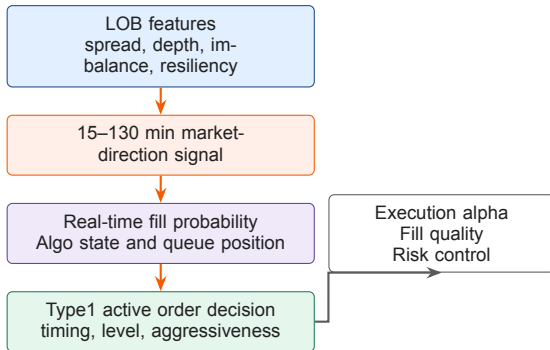
Feasibility status

- ▶ Research path and RMB 90mm base target are defined.
- ▶ Prototype can start with interpretable LOB features.
- ▶ Required data: book, order/fill, and internal flow timestamps.

Key risks to manage

- ▶ Data alignment across book, order, fill, and flow.
- ▶ Leakage control in labels and backtests.
- ▶ Cost attribution across direction, spread, impact, and routing.
- ▶ Regime stability in volatile or low-liquidity periods.

■ Algo Type1 Execution Alpha



Case 1: favorable direction

- ▶ Direction supports the order side.
- ▶ Expected fill probability is higher.
- ▶ Execution alpha: lower slippage and lower market impact.

Case 2: adverse direction

- ▶ Direction works against the Type1 order.
- ▶ Revise fill expectation and aggressiveness.
- ▶ Improve success rate and reduce unhedged risk.

■ Algo Plan and Literature Support

Implementation plan

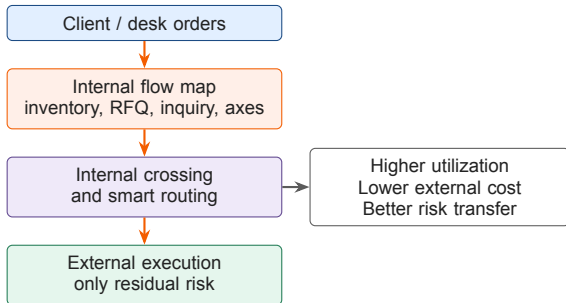
1. Build LOB feature stream by horizon: 15, 30, 60, 90, 130 minutes.
2. Start with transparent features: spread, depth, imbalance, queue depletion, resiliency.
3. Join direction signal with Algo fill-probability engine.
4. Run shadow trading, then controlled A/B test.

Why feasible

- ▶ Queue models link best-queue depletion to short-term price moves.
- ▶ Hawkes-style order-flow models support regime-dependent intensity signals.
- ▶ DeepLOB-style evidence shows book snapshots contain short-horizon structure.
- ▶ Microstructure benchmarks warn us to evaluate by cost-aware execution metrics, not accuracy alone.

Source: Cont and de Larrard; Bacry et al.; Zhang et al. DeepLOB; recent LOB benchmark studies.

■ Microstructure for Internal Flow



How microstructure helps

- ▶ Identify when external liquidity is expensive.
- ▶ Prioritize internal crossing before paying spread and impact.
- ▶ Route residual flow when book depth and resiliency are favorable.
- ▶ Feed realized cost back into desk-level flow intelligence.

■ How We Save 0.01 bp Per Trade

Base-case value

RMB 90mm

annual P&L target from 0.01 bp saving

- ▶ 2025 flow: RMB 45tn.
- ▶ Average duration: 2 years.
- ▶ $45tn \times 2 \times 0.01bp \approx RMB90mm$.

Where the saving comes from

- ▶ **Internal crossing:** avoid spread and external market impact.
- ▶ **Timing:** trade when depth and resiliency are favorable.
- ▶ **Adverse-selection control:** avoid passive fills when direction is unfavorable.

How we prove it

- ▶ Slippage vs. arrival / decision price.
- ▶ Market impact after execution.
- ▶ Internalization uplift and residual external cost.